

WEEK 3

Week 3: Node Provisioning - Lab Guide

HPC & Cluster Computing

Objective: Deploy a fully functional HPC provisioning system using the OpenHPC 4.1 methodology. **Goal**: Install Warewulf 4 on openEuler 24.03, import an openEuler container image, configure it automatically using OpenHPC scripts, and successfully boot a "diskless" compute node.

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1. Prerequisites

Ensure your Master Node has:

1. **Static IP**: `10.0.0.1` configured on your internal network adapter.
2. **Hostname**: `master`
3. **Internet Access**: Required to download the container image and packages.

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2. Environment Configuration (Crucial Step)

Explanation: This block loads the loc[illegible]es into the bash environment. These variables are subsequently substituted [illegible]guration files automatically, minimizing

```
source cluster_env.sh
```

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3. Configure Base OS

3.1 Configure Firewall

Warewulf relies on DHCP, TFTP, and HTTP. We will disable the firewall for this lab to avoid issues.

Explanation: This command stops and disables the default `firewalld` service so that Warewulf's internal TFTP/DHCP traffic is not blocked.

```
sudo systemctl disable --now firewalld || true
```

3.2 Configure NTP

HPC systems rely on synchronized clocks. We will configure Chrony.

Explanation: These commands install the `chrony` time service, enable it to start on boot, configure it to allow time queries from the local network (using the `ntp_server` variable we defined), and restart the service to apply changes.

```
sudo dnf -y install chrony
sudo systemctl enable chronyd.service
```

```
sudo echo "local stratum 10" >> /etc/chrony.conf
sudo echo "server ${NTP_SERVER}" >> /etc/chrony.conf
sudo echo "allow all" >> /etc/chrony.conf
sudo systemctl restart chronyd
```

4. Install OpenHPC Components

We will add the OpenHPC repository to the master node.

Explanation: This command fetches the official `ohpc-release` RPM, which configures the local `dnf` package manager to securely download software from the OpenHPC 4.1 community repository.

```
sudo dnf -y install http://repos.openhpc.community/OpenHP
```

5. Install Warewulf Provisioner

5.1 Install Packages and Create Directories

Install Warewulf and the `yq` utility (used for configuring YAML files).

Explanation: We install `warewulf-ohpc` which provides the provisioning framework, and `yq`, a command-line YAML processor. We also manually create the TFTP boot directory which Warewulf requires to serve bootloader files.

```
sudo dnf -y install warewulf-ohpc yq
sudo install -d -m 0755 /var/lib/tftpboot
```

5.2 Configure Warewulf (warewulf.conf & nodes.conf)

We use `yq` to automatically update the configuration based on the variables we exported earlier.

Explanation: Instead of using a text editor like `nano` or `vim`, we use `yq` to surgically modify the `warewulf.conf` YAML file using the environment variables defined in Section 2. We also use `sed` to update `nodes.conf` to mount `/opt` automatically. Finally, we enable the Warewulf service.

```
# Edit the warewulf.conf file to use appropriate interfac
sudo yq -i '.ipaddr = "'"${SMS_IP}"'"' /etc/warewulf/ware
sudo yq -i '.netmask = "'"${INTERNAL_NETMASK}"'"' /etc/wa
sudo yq -i '.network = "'"${INTERNAL_NETWORK}"'"' /etc/wa
sudo yq -i '.dhcp["range start"] = "'"${INTERNAL_NETWORK}
sudo yq -i '.dhcp["range end"] = "static"' /etc/warewulf/
sudo yq -i '.dhcp.template = "static"' /etc/warewulf/ware

# Edit the nodes.conf to mount /opt on boot
sudo sed -i "s/defaults,noauto,nofail,ro/defaults,nofail,

# Turn on debugging messages
sudo yq -i '.nodeprofiles.default.kernel.args -= ["quiet"
sudo echo "log-debug" >> /etc/dnsmasq.d/ww4-debug.conf

# Enable and start the warewulfd service before using wwc
sudo systemctl enable --now warewulfd
```

Explanation: These commands create a new Warewulf profile called "nodes" and a configuration overlay called "nodeconfig". The `wwctl profile set` commands attach the network variables to the profile so that any node using this profile inherits the correct network routing.

```
# Create a new "nodes" profile and inherit the "default"
sudo wwctl profile add nodes --profile default --comment

# Create a new "nodeconfig" overlay to store node configu
sudo wwctl overlay create nodeconfig
sudo wwctl profile set --yes nodes --system-overlays node

# Set default network configuration
sudo wwctl profile set -y nodes --netname=default --netde
sudo wwctl profile set -y nodes --netname=default --netma
sudo wwctl profile set -y nodes --netname=default --gatew
sudo wwctl profile set -y nodes --netname=default --netta

# Configuring Warewulf will restart/enable relevant servi
sudo wwctl configure --all
```

6. Create Compute Node Image

6.1 Import Base Image

We will pull the official openEuler image for Warewulf.

Explanation: This pulls a pre-built op container image from the GitHub Container Registry. Warewulf converts the into a bootable format. We also export the

```
sudo wwctl image import docker://ghcr.io/warewulf/warewul
export CHROOT=$(sudo wwctl image show openeuler-24.03)
```

6.2 Install Packages into the Image

We will execute a script inside the container to install necessary base utilities and the OpenHPC compute package.

Explanation: The `wwctl image exec --build=false` command runs commands *inside* the container image. The `<<- EOF` blocks pass a list of installation commands natively to the container's shell, automating the configuration without the need for manual interactive logins.

```
sudo wwctl image exec --build=false openeuler-24.03 -- /b
dnf -y install dnf-utils
dnf -y install http://repos.openhpc.community/OpenHPC/4/c
dnf -y update
EOF

sudo wwctl image exec --build=false openeuler-24.03 -- /b
dnf -y config-manager --add-repo https://eur.openeuler.op
dnf -y install openeuler-extra-repos
dnf -y install ohpc-base-compute
EOF
```

Set a root password for the image so you can log in later!

Explanation: Runs the `passwd` command inside the image so that the resulting compute node will have a known root password.

```
sudo wwctl image exec openeuler-24.03 /bin/passwd
```

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7. Deploy Cluster: Register Compute Nodes

We will configure our two compute nodes differently to demonstrate both provisioning models.

7.1 Stateless Node (compute-01)

This node will boot its OS entirely into RAM, leaving its local hard drive untouched. This is the default HPC behavior.

```
sudo wwctl node add --image=openeuler-24.03 --profile=noc
--ipaddr="${C_IP[0]}" --hwaddr="${C_MAC[0]}" "${C_NAME[0]
```

7.2 Stateful Node (compute-02)

This node will PXE boot, but Warewulf will install the OS permanently onto its local hard drive (`/dev/sda`). This is known as Stateful provisioning, which is useful for nodes that require large local storage or specific disk configurations.

Explanation: We configure the node and tell Warewulf's `wwinit` utility to wipe the specified disk and install the OS persistently.

```
sudo wwctl node add --image=openeuler-24.03 --profile=noc
--ipaddr="${C_IP[1]}" --hwaddr="${C_MAC[1]}" "${C_NAME[1]
```

```
sudo wwctl profile set nodes --init=wwinit
sudo wwctl node set "${C_NAME[1]}" --diskname=/dev/sda --
```

7.3 Finalize Configuration

Rebuild the container images and configuration overlays to apply all the changes.

```
# Rebuild image and overlays
sudo wwctl image build openeuler-24.03
sudo wwctl overlay build
```

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8. Boot the Nodes

1. Open the console/display of your **Compute Nodes (compute-01 and compute-02)**.
2. Ensure the BIOS/Boot Order on both VMs is set to **Network Boot / PXE** as the first priority.
3. Power on both nodes.
4. **Observe the Differences**:
   - **compute-01 (Stateless)**: Downloads the kernel and initramfs via HTTP and boots directly to the login prompt. A reboot resets the system entirely.
   - **compute-02 (Stateful)**: Downloads the kernel, executes `wwinit`, formats the virtual hard drive, writes the openEuler filesystem to the disk, and then pivots root. Subsequent local boots will run from the hard drive.



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9. Troubleshooting

If the node fails to boot:

1. **DHCP Timeout**: Could not get an IP.
   - Check `systemctl status warewulfd` on master.
   - Verify the MAC address in `wwctl node list node01` matches the actual VM MAC.
2. **TFTP Timeout**: Got IP, but stuck at "tftp...".
   - Firewall is usually the culprit (UDP port 69).

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10. Submission

Submit a PDF containing:

1. Screenshot of `wwctl image list` on the Master.
2. Screenshot of `wwctl node list` on the Master.
3. Screenshot of the **Compute Node console** showing a successful login prompt or successful boot log.

